

BANK TERM DEPOSIT SUBSCRIPTION ANALYSIS REPORT

This report presents an analysis of a bank marketing campaign dataset to identify factors influencing customer subscription to term deposits. The study includes exploratory data analysis, campaign effectiveness evaluation, correlation analysis, and predictive modeling using Logistic Regression.

1. Business Objective

- Identify customer characteristics that influence term deposit subscription.
- Evaluate effectiveness of marketing campaigns.
- Develop a predictive model to improve targeting strategy.

2. Dataset Overview

The dataset includes customer demographic details, campaign-related information, macroeconomic indicators, and the target variable (subscription: Yes/No).

The dataset is imbalanced:

- Approximately 88% customers did not subscribe.
- Approximately 12% customers subscribed.

3. Exploratory Data Analysis

AGE DISTRIBUTION

- Customers aged 60+ show the highest subscription rate.
- Middle age groups (31–50) show relatively lower conversion rates.

JOB DISTRIBUTION

- Students and retired customers have higher subscription likelihood.
- Blue-collar and services sectors show lower conversion rates.

CONTACT METHOD ANALYSIS

- Cellular contact performs significantly better than telephone contact.
- Modern communication channels improve customer engagement.

CAMPAIGN INTENSITY

- Customers contacted only once show higher conversion rates.
- Subscription probability decreases as number of contacts increases.
- Over-contacting reduces campaign effectiveness.

PREVIOUS CAMPAIGN OUTCOME

- Customers with previous campaign success have very high probability of subscribing again.
- This is one of the strongest predictive indicators.

4. Correlation Analysis

- Call duration positively correlates with subscription.
- Euribor rate negatively correlates with subscription.
- Strong multicollinearity observed among macroeconomic variables (employment variation rate, Euribor rate, number of employees).

5. Predictive Modeling – Logistic Regression

Model Improvements:

- Feature scaling applied.
- Class imbalance handled using `class_weight='balanced'`.

Final Model Performance:

- Accuracy: ~82%
- Recall (Subscribers): 64%
- F1-score (Subscribers): 0.45

The balanced model significantly improved detection of actual subscribers compared to the baseline model.

6. Key Business Insights

- Older customers (60+) are highly responsive.
- Students and retired individuals are strong target segments.
- Cellular communication is more effective than telephone.
- Fewer contacts increase conversion probability.
- Previous campaign success strongly influences future subscription.
- Macroeconomic indicators impact customer decisions.

7. Recommendations

1. Prioritize customers aged 60+ and students.
2. Focus marketing efforts on cellular outreach.
3. Avoid excessive follow-up calls.
4. Retarget customers with previous successful campaign outcomes.
5. Deploy predictive model for targeted marketing to reduce campaign cost.